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GE 4120: ELECTIVES

**Easing the Wait: An IT-Enabled Patient Queuing and Scheduling System
for Reducing Long Waiting Times in Hospitals and Clinics.**

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I. Introduction

At present, hospitals and clinics serve as the frontline of healthcare delivery, yet the very process of receiving care often begins with a long and tiring wait. Patients arriving for a simple check-up or a scheduled consultation frequently find themselves spending hours in crowded waiting areas before they are even attended to, whether at the registration counter, the consultation room, the laboratory, or the pharmacy. While healthcare facilities strive to serve every patient who walks through their doors, the reliance on manual, paper-based processes and the sheer volume of patients seeking care each day often overwhelm the limited number of staff and service windows available. In this sense, this results in overcrowded waiting rooms and prolonged patient dwell times, creating a strain that affects not only the comfort of patients but also the efficiency of hospital operations and, in more serious cases, the timeliness of the medical attention they receive. This particular problem of long waiting times relates to a modern challenge faced by healthcare systems today, both in large hospitals and small community clinics, and is important to address by providing an ideal, technology-driven solution that can ease congestion and improve the overall patient experience. In this proposal, the author seeks to detail and scope the problem of long waiting times in hospitals and clinics, and to outline an ideal information technology solution that automates queuing and scheduling in order to reduce the time patients spend waiting for care.

Problem description

In modern hospital and clinic settings, long waiting times can occur at nearly every stage of a patient's visit, not just at the doctor's consultation, but also during registration, diagnostic testing, and medication release. Front-desk staff who rely on manual queuing methods, such as handwritten number slips or verbal call-outs, often struggle to track patient order accurately, especially when the number of walk-in patients rises sharply during peak hours. With this, the lasting impact of long waiting times on patients affects not only their comfort but also their trust in the healthcare system, leading to frustration, missed follow-up visits, and, in more urgent cases, delayed treatment that can worsen a patient's condition. According to Zhang et al. (2023), waiting time has a measurable effect on patient satisfaction in outpatient departments, with longer waits consistently associated with lower satisfaction scores regardless of the quality of clinical care received. Attending to a high volume of patients helps hospitals fulfill their duty of

providing accessible healthcare; however, when patient volume exceeds what manual systems can efficiently manage, the result is overwhelming congestion, not only for staff trying to coordinate patient flow but also for patients waiting for care, increasing the risk of delayed diagnosis and treatment. Patients who arrive without a scheduled appointment are especially affected, as they are placed into general queues that are not properly prioritized or balanced against the day's existing bookings, leading to unpredictable and often lengthy waits. With this in mind, long waiting times result in overcrowded lobbies, rising stress among patients and staff alike, and a general decline in the quality of the overall healthcare experience.

With this observation in mind, it is worth noting how long waiting times can affect the operations of hospitals and clinics on a small scale, such as a single outpatient department during a busy morning. However, the problem is not limited to one department or facility; it reflects a broader, recurring pattern across many healthcare institutions that still depend on manual queuing and scheduling. Some of the underlying issues are limited to the queuing process itself, but they also extend to how appointments are scheduled, how walk-in patients are integrated into existing queues, and how information about wait times is, or is not, communicated to patients. Some challenges identified in this problem are particularly relevant to high-volume public hospitals and community clinics, as this often does not reflect how private or smaller facilities experience the same issue, or whether the severity differs by patient volume and available staff.

With this in mind, the proposal of an ideal solution can be mitigated by using a digital patient queuing and scheduling system that automates registration, appointment booking, and real-time queue tracking.

Proposal solution.

The proposed solution to mitigate long waiting times among patients in hospitals and clinics is the development of MediQueue, an automated patient queuing and scheduling system integrated with real-time notifications. This system is designed to help patients and hospital staff systematically manage, organize, and monitor patient flow from arrival to service completion through digital check-in and structured appointment scheduling. By using an online booking portal and a self-service kiosk or QR code check-in, patients can reserve consultation slots in advance or register upon arrival without relying on manual number slips, enabling hospital staff

to visualize patient queues clearly and prevent bottlenecks at the registration counter. One of the key features of the system is the smart scheduling and load-balancing module, which distributes appointments across available time slots and prioritizes urgent cases to reduce overcrowding during peak hours. Additional features include a centralized queue management dashboard for staff, real-time digital display boards and SMS or app notifications for patients, and an analytics and reporting module that tracks average wait times and peak-hour patterns.

The system primarily targets public and private hospitals, outpatient clinics, and community health centers, where high patient volume and limited administrative staff are common. By automating queue management and appointment scheduling, the system encourages more organized patient flow, minimizes overcrowding in waiting areas, and prevents the prolonged, uncertain waits that contribute to patient dissatisfaction. The expected impact of this solution includes reduced average waiting time at registration, consultation, and pharmacy service points, improved staff productivity through automated patient routing, better data-driven decision-making for hospital administrators, and a more comfortable, transparent waiting experience for patients. Overall, integrating digital queuing and scheduling offers a sustainable, scalable approach to reducing long waiting times in hospitals and clinics of varying sizes.

II. Conclusion

Long waiting times caused by manual queuing, high patient volume, and inefficient scheduling have become a prevalent issue affecting hospitals and clinics alike, leading to negative consequences such as patient dissatisfaction, staff strain, and, in urgent cases, delayed access to care. Addressing this problem requires not only awareness but also structured and practical technology solutions. This proposal highlights the importance of implementing an automated patient queuing and scheduling system as a sustainable approach to managing patient flow and reducing waiting times. By digitizing registration, organizing appointments, and providing real-time queue visibility, the proposed system supports faster, more organized, and more patient-centered healthcare delivery. Ultimately, adopting systematic, technology-driven queuing and scheduling practices can help healthcare facilities effectively cope with high patient volume, improve service efficiency, and foster long-term satisfaction for both patients and staff.

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